SYSTEM DESIGN · OCTOBER 2026

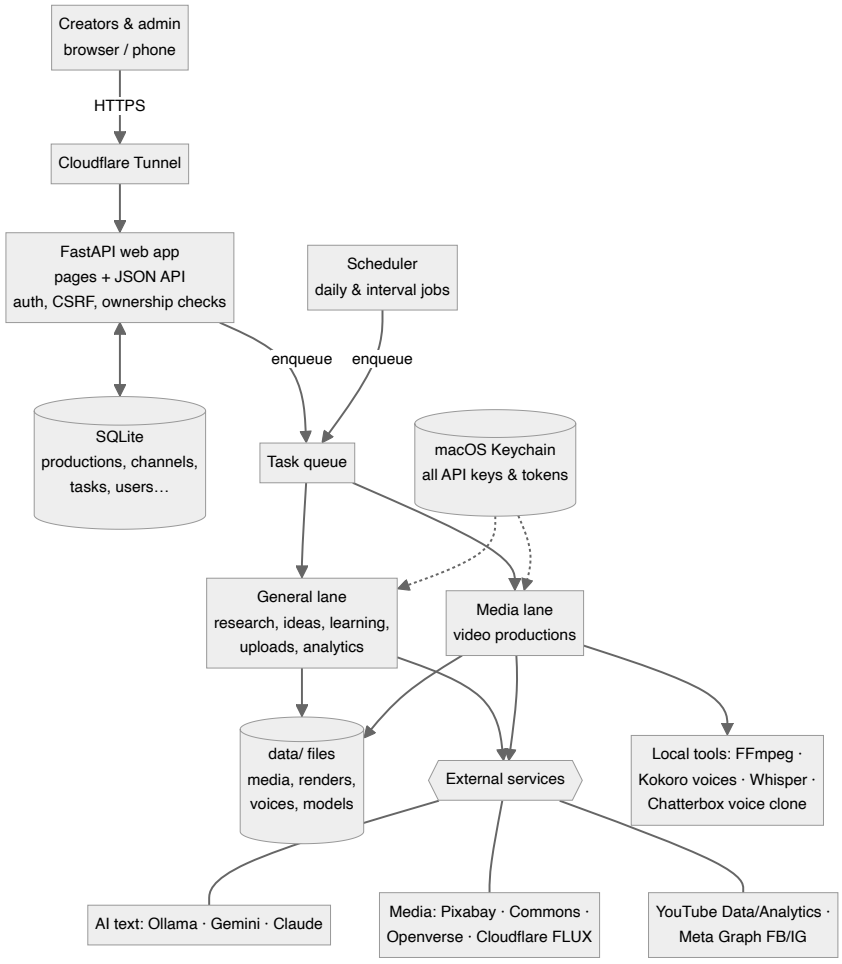
AI Video Studio: how the whole platform works

A multi-user web studio that turns an idea, a prompt, a full screenplay or an uploaded video into a finished YouTube-ready video, then publishes it on schedule to YouTube, Facebook and Instagram. It runs on one Mac, is reachable publicly through a Cloudflare Tunnel, and uses free tools first.

Python 3.12 · FastAPI + Jinja **SQLite** · 16 migrations **2** worker lanes
6 video workflows **~127** web routes **250+** automated tests

System architecture

One process serves the website and runs the background workers and the scheduler. Long work (research, rendering, uploads) never runs inside a web request; it is queued as a task.



Web layer

`studio/web/app.py`: every page and API route, sign-in, per-session CSRF tokens, and a middleware that checks each URL belongs to the signed-in workspace.

Worker

`studio/worker.py`: claims tasks from SQLite, retries with backoff, and treats temporary problems (YouTube limits, network drops, free-allowance waits) as "try later" without using up attempts.

Pipelines

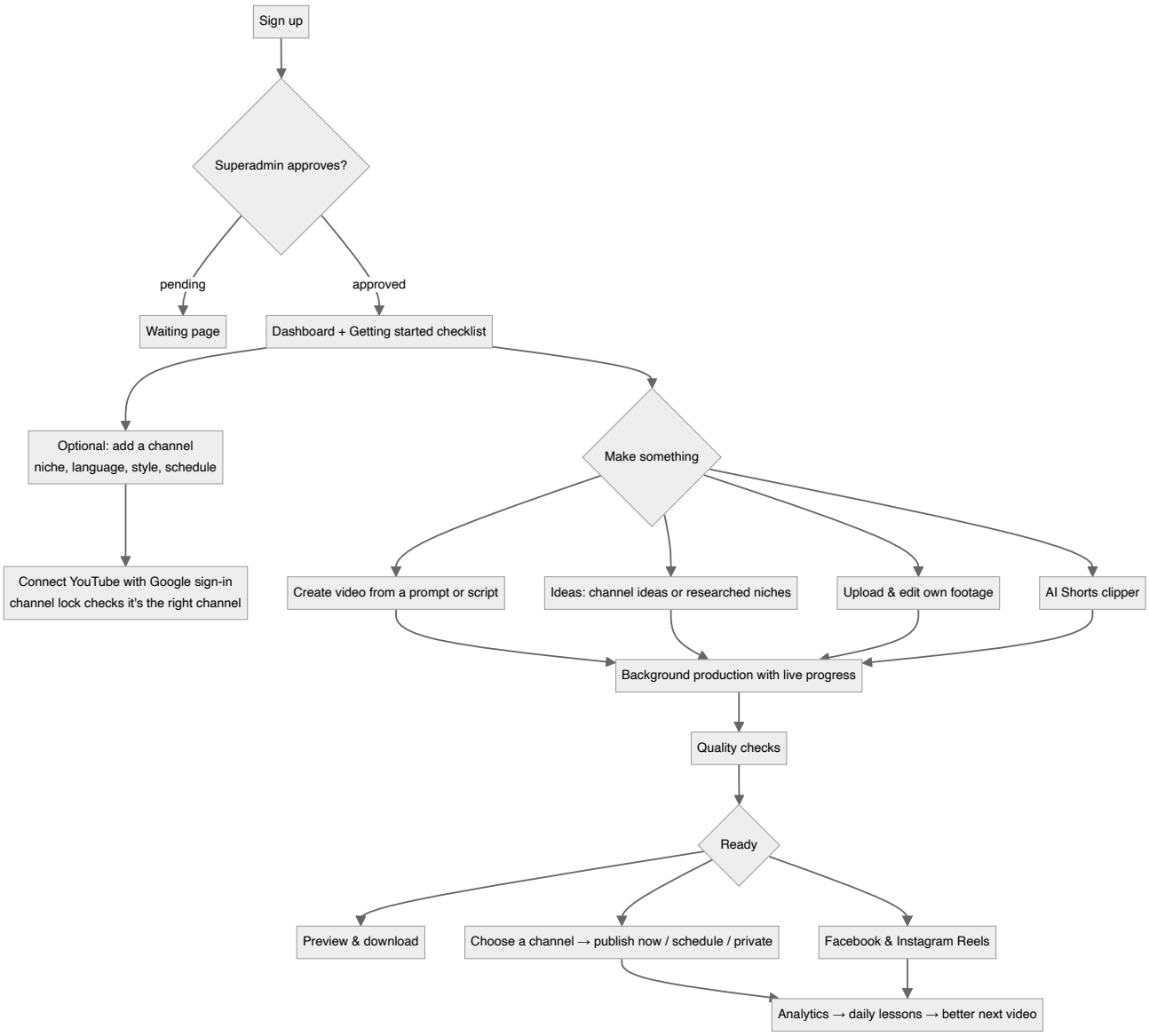
`pipeline.py` runs each video as a list of resumable stages; each workflow (researched, story, film, edit, finance) supplies its own stages.

Secrets

Keys live only in the macOS Keychain (service `yt-automation-studio`), never in the database, files or GitHub.

User journey

What a creator does, from first visit to published video.

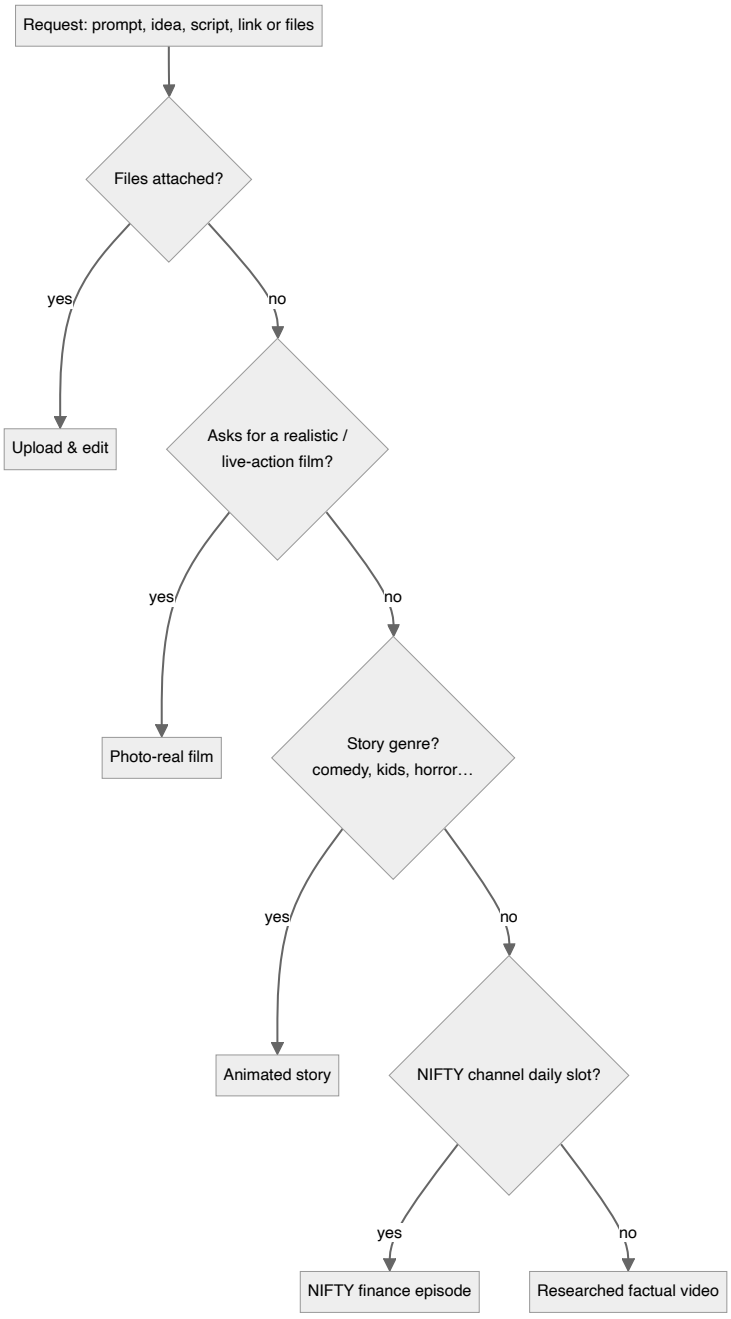


Videos made from your own prompt or script start with **no channel**. You decide where (or whether) to publish after watching it. Channel characters and channel style are only used when a video is made for that channel.

Video workflows

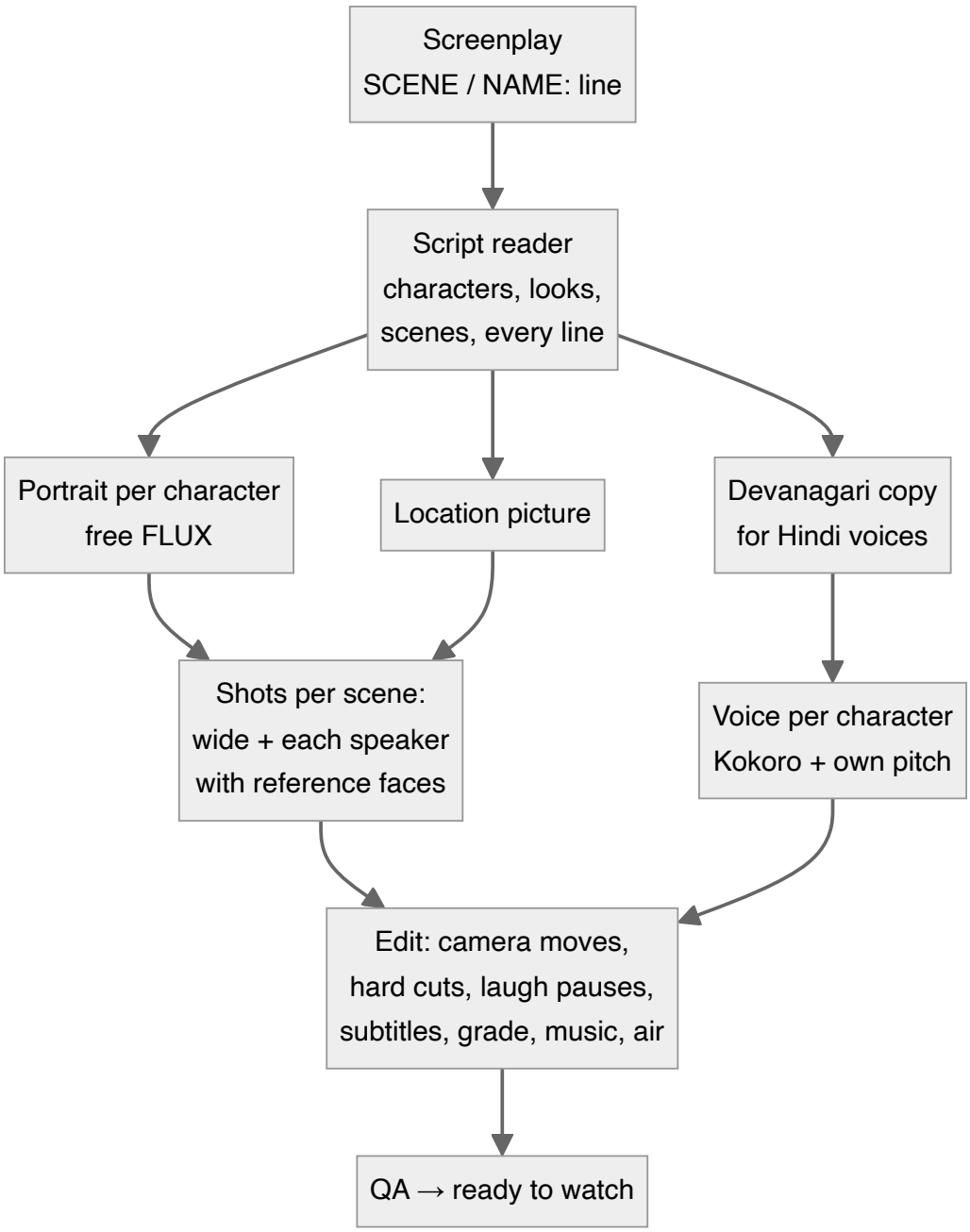
Every video is a `production` row with a `kind`. Each kind runs its own stages; a stage that is interrupted resumes where it stopped.

How a request is routed



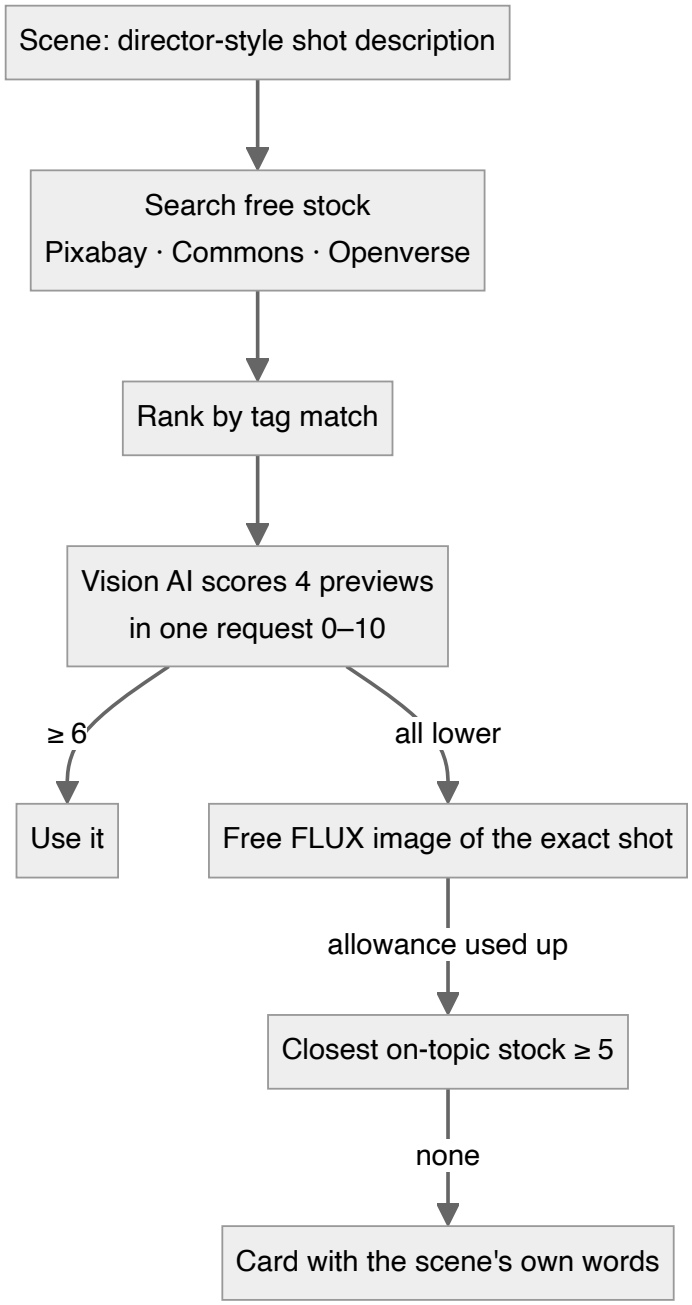
WORKFLOW	STAGES	OUTPUT
Researched factual `general`	research → script → review → assets → voice → render → thumbnail → QA	Long video and/or Shorts with sourced facts, captions, chapters, credits
Animated story `story`	script → characters → voice → animate → thumbnail → QA → review (+ Shorts cut from the episode)	Cartoon episode with a recurring or one-off cast, per-character voices, lip movement from audio
Photo-real film `film`	script (read as written) → cast portraits + location → voices → shots → edit → QA	Realistic-looking film: consistent faces from reference portraits, a distinct voice per character, subtitles, film grade
Upload & edit `edit`	inspect → transcribe (Whisper) → AI edit plan → render → review	Edited main video plus Shorts, optional re-voice in your cloned voice
AI Shorts clipper	find best moments → vertical reframe → captions	Batch of Shorts from a long video (never auto-published)
NIFTY finance `nifty_pivot`, `fib_lesson`	verified market data → script → compliance gate → render → upload	Daily levels episode and Shorts; blocked if data isn't verified or wording breaks the rules

Photo-real film in detail



Visuals & style rules

Every scene's picture must match what is being said, and each channel keeps its own look.



Kids channels

Cartoon and illustration stock plus cartoon-style FLUX images. Photo-only sources are skipped and real photos score 0–2.

Market & factual

Real footage and photo-real images only, never cartoons, with a cinematic grade (softer colour, warm highlights, light grain).

Stand-alone videos

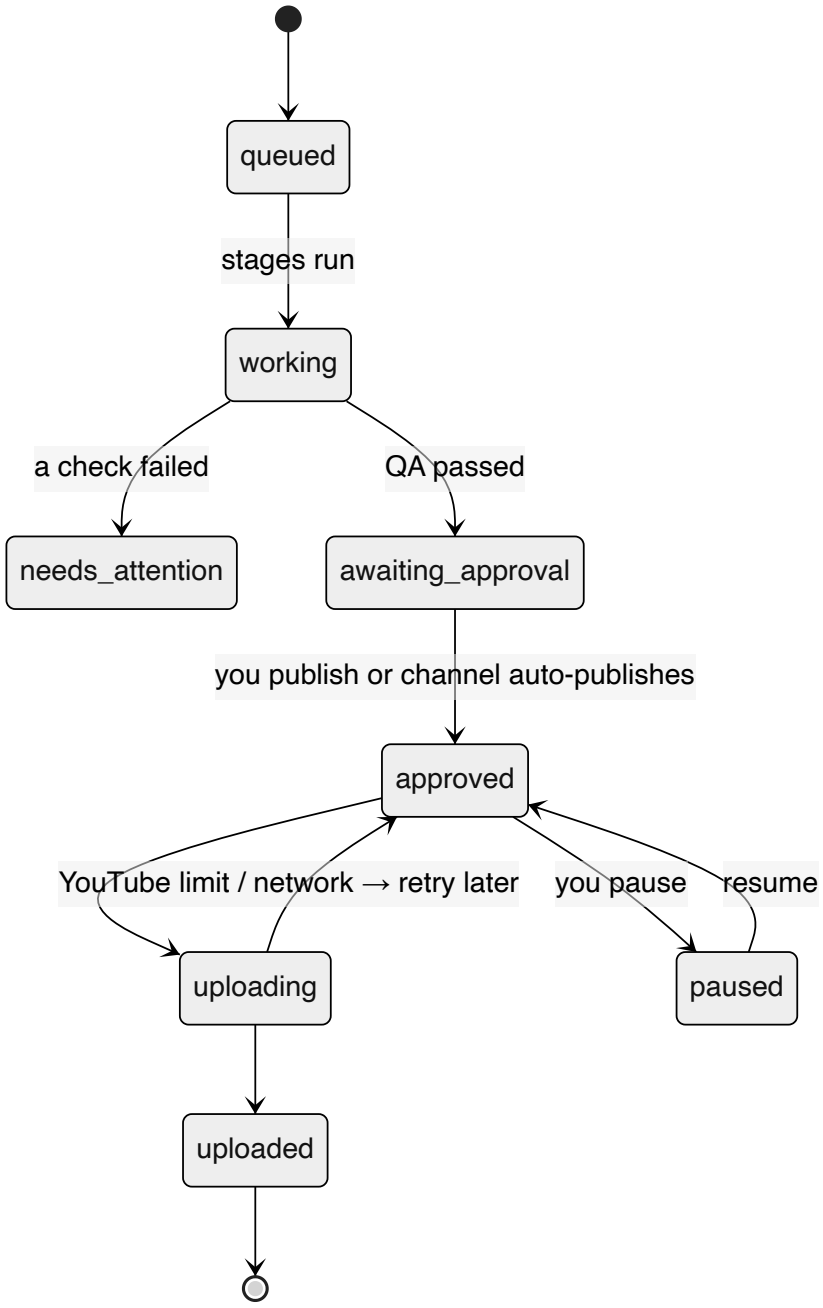
Style comes from the video's own topic. They're kept in a hidden download-only holder, never filed under a real channel.

AI & free-first providers

Free options are always tried first; paid ones only run when every free one fails, and only within monthly caps.

JOB	TRIED IN ORDER
Scripts, ideas, research, learning	FREE Ollama on this Mac → Ollama Cloud → Gemini free tier → PAID Claude
Picture checks	FREE Gemini Flash-Lite (paced to the free limit) → Ollama Cloud vision
Scene images	FREE Cloudflare FLUX (~10,000 neurons/day) → LOCAL Stable Diffusion; paid Gemini images switched off
Narration voices	LOCAL Kokoro (Hindi + English) · PAID Sarvam when credits exist, falls back to Kokoro automatically · LOCAL your cloned voice (Chatterbox, with consent)
Speech-to-text	LOCAL Whisper small, offline
AI video clips	PAID Gemini Omni, opt-in per job with a cost estimate and monthly cap

Publishing & schedules



1. **One main video per daily slot.** A backlog spreads over the following days; Shorts share their episode's day and never go public before it.
2. **A passed slot moves forward.** If a video waited (limit, pause), it takes the channel's next free slot instead of being refused.
3. **Temporary problems retry forever.** YouTube's daily limit, quota, outages and dropped connections never use up attempts.
4. **Dashboard "Publishing needs you".** Paused videos, held Shorts, the daily limit and a deleted scheduled video each get a one-click fix.

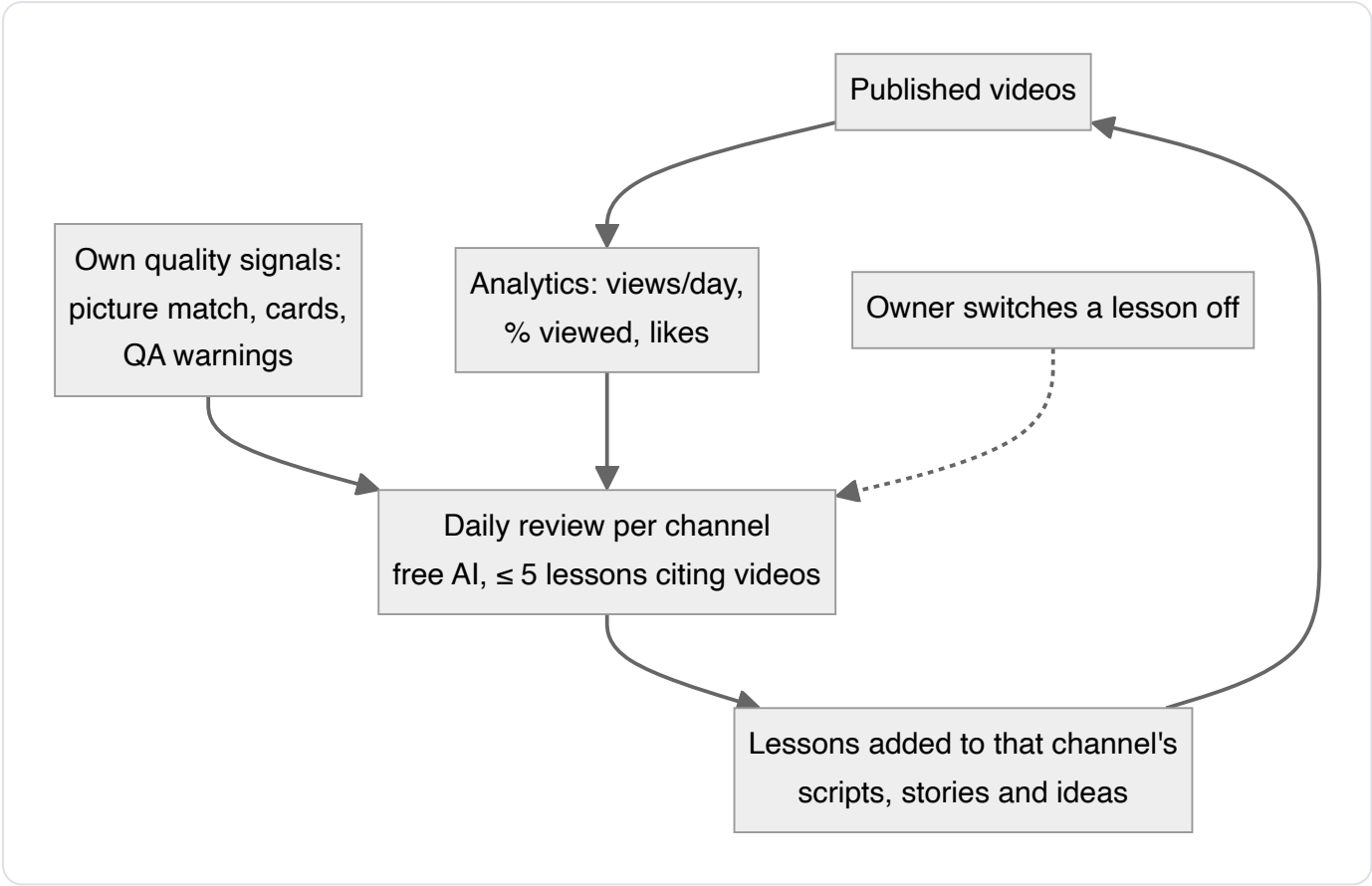
5. **Facebook & Instagram** via Meta Graph API: Reels and Page videos, one tracked post per platform, never posted twice.

Daily automation (IST)

TIME	JOB	WHAT IT DOES
03:30	Housekeeping	Clears old working files
06:00	Channel growth	Subscribers, watch hours, monetization progress
06:30	Learn from results	Writes each channel's lessons for its next videos
every 6 h	Analytics sync	Views, watch time, % viewed per video
15:40 Mon–Fri	NIFTY daily episode	Only when the day's verified market data is imported
16:00	Fibonacci lesson series	One part a day, 7 parts
per channel	Channel daily video	Each channel's own time (e.g. Kids at 18:00)

Missed runs (Mac asleep, app closed) run once on wake. Uploads retry automatically.

Learning loop



Audience lessons need at least 3 public videos older than 2 days. Stand-alone videos never pick up or feed a channel's lessons.

Data model

TABLE	HOLDS
`workspaces`, `users`, `workspace_members`, `sessions`	Accounts, plan, feature permissions, approval status, hashed session tokens
`channels`	Niche, language, workflow, schedule, publish time, upload switch, YouTube link, Meta Page, kids flag; hidden stand-alone holder
`productions`	Every video: kind, stage, status, script package, timeline, QA, files, publish time, YouTube id
`topics`, `niches`, `evidence`	Ideas, researched niches and cited sources
`assets`	Every picture/clip used, its licence, credit and why it matched (vision score)
`characters`, `voices`	Story cast bible; cloned voices with recorded consent
`tasks`, `schedules`	Background queue with retries; recurring jobs
`analytics`, `lessons`	Per-video results; what each channel learned
`social_posts`, `media`	Facebook/Instagram posts; uploaded files for editing
`ai_usage`, `usage`	Every AI call and its cost; plan metering

Accounts & security

Multi-tenant

Every channel, video, idea, voice and file belongs to one workspace; URL ownership is checked on every request, including media files.

Superadmin control

New sign-ups wait for approval; plans (Free, Creator, Pro, Studio) and per-account feature switches.

Web protections

scrypt passwords, hashed sessions, CSRF tokens, strict Content-Security-Policy, safe redirects, sign-in rate limits.

Publishing safety

Channel lock refuses uploads to the wrong YouTube channel; NIFTY compliance gate; finance wording rule only on finance videos.

Hosting & setup

1. Runs as a macOS LaunchAgent on port 8765; installers exist for Mac and Windows.
2. Public access through a Cloudflare Tunnel on your own domain.
3. **Computer & setup** page detects Apple Silicon / Intel / AMD and the GPU, lists missing parts and installs Kokoro, Whisper and voice cloning in one click.
4. Keys are added under Settings → Credentials and stored in the Keychain.
5. A separate test instance (port 8790, own data folder) is used for trying changes without touching real channels.

Known limits

AREA	LIMIT TODAY	WAY FORWARD
Realistic acting	Free films use still shots; people don't move or speak on screen	Paid AI video (about $0.10/s at 720p) for key shots
Animated quality	Code-drawn cartoons look simple	Illustrated character art pipeline (in progress)
Free image speed	About 1 minute per FLUX image; ~60–90 images a day	Long films finish over hours or the next day automatically
Hindi voices	Two free male and two female Kokoro voices	Pitch variants, Sarvam credits, or cloned voices
Voice cloning speed	About 8–10× slower than real time on an M2	Use for Shorts or narration only
Already-uploaded videos	App's YouTube access can upload but not edit	Edits are made in YouTube Studio

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Written from the current codebase (`studio/`) and live configuration on 29 October 2026.